

- (1) What are the parameters of your application ?

An application has three parameters that determine a location in HDFS of the files with the application processing times.

- (2) What information is filtered out by a mapper ?

The mapper reads all three files and filters out nothing.

- (3) What key-value pairs are created by a mapper ?

A mapper uses the contents of all three files to create key-value pairs where key = customer# and value = <region, total_time>. Each row from the files contributes to one key-value pair.

- (4) How key-value pairs are created by a mapper ?

A mapper reads the files row by row and creates one key-value pair from each row.

- (5) What key-value pairs are processed by a reducer ?

customer# - [<region, total_time>, ... , <region, total_time>]

- (6) How key-value pairs are processed by a reducer ?

For a given customer# a reducer create a 3 bits map [bit_a,bit_b,bit_c] and sets the map to [0,0,0] and initialize total to 0.

Then, a reducer iterates over a list of values [<region, total_time>, ... , <region, total_time>]. If region=a then bit_a is set to 1, if region=b then bit_b is set to 1, if region=c then bit_c is set to 1, total is increased by total_time.

When the iterations for a given customer# are completed and all 3 bits are set to 1 a reducer displays customer# and total. Otherwise, it skips the current customer and takes the next one.

- (7) What is the format of the final results ?

<customer#, total time>

- (8) How to upload data to HDFS ?

```
hdfs dfs -put region_a.txt .  
hdfs dfs -put region_b.txt .  
hdfs dfs -put region_c.txt .
```

- (9) How to prepare your application for processing ?

```
chmod u+x mapper.py  
chmod u+x reducer.py
```

(10)How to list the final results ?

```
hdfs dfs -cat output/p*
```